SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct: (192526013)

I Veebala Jeevan *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Jeevan

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	-	1	1
Debugging	25%	2	1	2	2	2
Optimization	20%	-	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

25/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COMPUTER NETWORKS

COURSE CODE: CSA07

DEBUGGING & OPTIMISATION TASK

COURSE OUTCOME COVERED	CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology (BL3).
ASSESSMENT TOOL WEIGHTAGE	30%
SOFTWARE	Cisco Packet Tracer and Wireshark

Student Details

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Date of Submission	____29/10/26_____

Code of Conduct

I ___veerala jeevan (1925260133)_____ (Name / Register Number) certify that this submission is my original workand that I have adhered to the guidelines specified for this assessment. I understand that any violation of academicintegrity rules will result in disciplinary action.
Signature of the student: ____veerala jeevan _____

Introduction and Methodology

This assignment addresses the five debugging and optimisation tasks in Assessment Tool 3 for Computer Networks (CSA07). The work focuses on IPv4 subnet boundaries, IPv6 prefix interpretation, Variable Length Subnet Masking (VLSM), and router forwarding decisions. Each problem is approached by checking the supplied configuration, calculating the relevant network properties, identifying what the evidence does and does not prove, and proposing a corrected or more efficient design.

A key principle in network troubleshooting is to separate a confirmed addressing error from a possible cause. An address outside the intended subnet is a clear configuration defect. By contrast, two IPv6 addresses can be in the same /64 even when their written forms look different; in that case, the prefix alone does not explain a failure, and link, routing, firewall, neighbor discovery, or duplicate-address issues must also be checked.

General Debugging Workflow

- Record the address, prefix or mask, gateway, destination, and expected network.
- Calculate network boundaries before changing any settings.
- Check whether each address is valid for its intended subnet and whether a gateway is reachable on the local link.
- Make the smallest correction that matches the stated design requirement.
- Verify the result using host configuration commands, ping where permitted, route inspection, and packet capture.

Tools and Verification

Cisco Packet Tracer can be used to build a small topology, assign addresses, configure routers, inspect routing tables, and test connectivity with ping and simulation mode. Wireshark can capture ARP, ICMP, ICMPv6 Neighbor Solicitation/Advertisement, and other relevant packets. A successful ping is useful evidence of reachability, while a failed ping should be interpreted alongside interface status, routes, filtering rules, and packet traces.

Important assumptions

The source assessment does not provide the actual host counts for the departments in Question 3, nor a complete physical topology or router interface configuration for Questions 2 and 5. Where these details are missing, the answer states the limitation and uses a clearly labelled example or verification plan instead of claiming that an unspecified fact is known.

Question 1 – Debugging an IPv4 Address and Gateway

Given configuration: Workstation IP address 192.168.10.65, subnet mask 255.255.255.192 (/26), and default gateway 192.168.10.1. The intended subnet is stated as 192.168.10.0/26.

1.1 Identify the subnet

A /26 mask has 26 network bits and 6 host bits. In dotted-decimal notation, 255.255.255.192 means that the last octet is divided into blocks of 64 addresses. The subnet boundaries in the 192.168.10.x range are .0, .64, .128, and .192. The configured host address ends in .65, so it belongs to the block that starts at .64, not the block that starts at .0.

Property	Correct value for 192.168.10.65/26
Network address	192.168.10.64
Broadcast address	192.168.10.127
Valid host range	192.168.10.65 – 192.168.10.126
Subnet mask	255.255.255.192
Total addresses	64
Usable host addresses	62

1.2 Verify the configured IP and gateway

The workstation address 192.168.10.65 is a valid host address within 192.168.10.64/26. It is neither the network address (.64) nor the broadcast address (.127). The problem is the gateway: 192.168.10.1 belongs to 192.168.10.0/26, whose host range is 192.168.10.1 through 192.168.10.62. Therefore, the workstation and its configured gateway are in different /26 subnets.

A host normally needs a reachable router interface on its local subnet as the default gateway. With this mask, the workstation cannot treat 192.168.10.1 as a directly connected neighbor. Unless a special design such as proxy ARP or additional routing is intentionally configured, the gateway setting is invalid for this workstation's subnet.

1.3 Root cause and correction

The assessment says that the workstation should communicate with systems in 192.168.10.0/26. The simplest correction is to place the workstation in that intended subnet and retain 192.168.10.1 as the gateway.

Setting	Current value	Recommended value
IP address	192.168.10.65	192.168.10.2 (example valid host)
Subnet mask	255.255.255.192	255.255.255.192
Default gateway	192.168.10.1	192.168.10.1

The selected address 192.168.10.2 is an example; any unused address from 192.168.10.2 to 192.168.10.62 is suitable, provided it is not already assigned and the gateway router interface is configured as 192.168.10.1/26.

Question 1 – Validation and Host Capacity

1.4 Network calculation

The subnet mask 255.255.255.192 corresponds to binary 11111111.11111111.11111111.11000000. The last six bits are host bits, so the number of addresses is 2■ = 64. In a conventional IPv4 subnet, the network and broadcast addresses are reserved, leaving 64 – 2 = 62 usable host addresses.

Address block	Network address	First host	Last host	Broadcast
192.168.10.0/26	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
192.168.10.64/26	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127

1.5 Step-by-step debugging in Packet Tracer

- Open the workstation's IP Configuration window and verify the IPv4 address, mask, and gateway.
- Check the router interface connected to the 192.168.10.0/26 LAN. Its address should be in the same subnet, for example 192.168.10.1 with mask 255.255.255.192.
- Set the workstation to an unused address in 192.168.10.1–192.168.10.62. If .1 is the router, choose a different host such as .2.
- Ping 192.168.10.1 first. If it fails, inspect cabling, interface state, VLAN membership, and address configuration.
- Ping another host in 192.168.10.0/26. If local gateway ping works but the remote host fails, check the other host's IP/mask, duplicate addresses, and any host firewall.
- Use Packet Tracer Simulation mode or Wireshark to observe ARP requests and ICMP replies. Repeated ARP requests without a reply can indicate that the target is not reachable at Layer 2 or that its address is wrong.

1.6 Optimisation and conclusion

The corrected configuration aligns the workstation with the intended network and avoids unnecessary routing complexity. The /26 subnet still provides 62 usable host addresses; changing the workstation address does not change the subnet's capacity. Address allocation should be documented and preferably managed through a consistent static-address plan or DHCP reservations to avoid duplicate IP addresses.

Answer summary: 192.168.10.65/26 belongs to 192.168.10.64/26. The current IP is valid for that subnet, but the gateway 192.168.10.1 is not in it. To communicate with 192.168.10.0/26, use an unused host address in that subnet, for example 192.168.10.2/26, with gateway 192.168.10.1. The corrected subnet has 62 usable IPv4 host addresses.

Question 2 – Debugging the IPv6 Addressing

Given: Device A is configured as 2001:db8::ff00:42:8329/64 and Device B as 2001:db8:0:0:1::1/64. The assessment states that they are connected to the same network but communication fails.

2.1 Expand the IPv6 addresses

IPv6 addresses contain eight 16-bit groups (hextets). The double colon (::) represents one or more consecutive groups of zero and can appear only once in an address. Each expanded group is written as four hexadecimal digits.

Device	Given address	Expanded address
A	2001:db8::ff00:42:8329/64	2001:0db8:0000:0000:0000:ff00:0042:8329/64
B	2001:db8:0:0:1::1/64	2001:0db8:0000:0000:0001:0000:0000:0001/64

2.2 Determine each /64 network prefix

A /64 prefix identifies the first four hextets (the first 64 bits) as the network portion. The remaining four hextets form the interface identifier portion. For Device A, the first four hextets are 2001:0db8:0000:0000. For Device B, they are also 2001:0db8:0000:0000. Therefore both addresses belong to the same /64 prefix: 2001:db8:0:0::/64.

Device	First 64 bits (prefix)	Interface identifier
A	2001:db8:0:0::/64	0000:ff00:0042:8329
B	2001:db8:0:0::/64	0001:0000:0000:0001

2.3 Is there a subnet mismatch?

No. Although Device B contains a 1 in the fifth hextet, that hextet is after the first four hextets and therefore lies in the interface identifier for a /64. It does not change the /64 network prefix. A common debugging mistake is to compare the first five groups visually and incorrectly conclude that the devices are on different subnets.

The supplied addresses alone do not prove an IPv6 prefix mismatch. If both devices are truly on the same Layer 2 network and both use /64, their IPv6 prefixes are consistent. The communication failure must be investigated through additional checks, including whether addresses are actually assigned, interfaces are up, Neighbor Discovery succeeds, ICMPv6 is permitted, and the devices are attached to the expected VLAN or link.

Question 2 – IPv6 Optimisation and Verification

2.4 Appropriate addressing

Because both supplied addresses are already within the same /64, it is not necessary to change either address merely to make their prefixes match. A suitable corrected configuration can retain the original addresses and ensure that both devices use prefix length /64 on the same link. For clarity, a simpler illustrative pair could be Device A 2001:db8:0:0::10/64 and Device B 2001:db8:0:0::20/64, provided those addresses are unused and the network is designed to use 2001:db8:0:0::/64.

Parameter	Device A	Device B
IPv6 address (original)	2001:db8::ff00:42:8329	2001:db8:0:0:1::1
Prefix length	/64	/64
Network prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Same /64?	Yes	Yes

2.5 Available address count

A /64 leaves 64 bits for the interface identifier. The number of bit patterns is 2■■ = 18,446,744,073,709,551,616 possible interface-identifier values. Unlike ordinary IPv4 subnet calculations, IPv6 does not use a broadcast address, and there is no universal rule that subtracts two addresses to find a standard usable-host count. Some interface identifier values have special meanings or are reserved by particular mechanisms, so the practical assignment policy should follow the IPv6 design and operating-system requirements.

2.6 Troubleshooting procedure

- Confirm the exact IPv6 address and prefix length on both devices. Verify that both are configured with /64 and that the addresses are not tentative or duplicated.
- Check that both interfaces are up and connected to the intended same VLAN or IPv6 link.
- Test local IPv6 reachability with ping to the peer's full address, where ICMPv6 echo is allowed.
- Inspect Neighbor Discovery. A device should resolve the peer's link-layer address using ICMPv6 Neighbor Solicitation and Advertisement messages.
- Check host and network firewall rules for ICMPv6. Blocking essential ICMPv6 traffic can disrupt more than ping, including Neighbor Discovery and Path MTU Discovery.
- Review router and switch configuration if the devices are not actually on the same Layer 2 segment. Confirm VLANs, router interfaces, and any required routes.
- Use Wireshark with an IPv6 display filter such as `ipv6` or `icmpv6` to determine whether packets leave the sender, whether Neighbor Discovery receives a reply, and whether responses return.

2.7 Conclusion

The important result is that the two given addresses share the same /64 prefix. The assessment describes a failure, but the provided addressing information does not identify a prefix mismatch. The correct debugging response is to preserve the valid prefix configuration and inspect Neighbor Discovery, interface/link status, filtering, duplicate addresses, and the actual topology before changing addresses.

Question 3 – Optimising IPv4 Subnet Allocation with VLSM

The organisation has 192.168.1.0/24 and has used /26, /27, and /28 masks for different departments. Some departments report insufficient addresses while other departments have unused addresses. The assessment does not specify department names, host counts, or the exact current subnet allocations. Therefore, an exact audit of the existing plan cannot be calculated from the prompt alone. The following explains the subnet sizes and provides a worked VLSM design using clearly stated example requirements.

3.1 Capacity of the given masks

Prefix	Mask	Total addresses	Usable IPv4 hosts	Block size
/26	255.255.255.192	64	62	64
/27	255.255.255.224	32	30	32
/28	255.255.255.240	16	14	16

A department with more than 62 hosts cannot fit into a /26; more than 30 hosts cannot fit into a /27; and more than 14 hosts cannot fit into a /28. Conversely, assigning a /26 to a department that needs only 12 devices wastes a large number of addresses. The correct mask depends on the maximum expected number of interfaces, not simply the number of staff members.

3.2 Subnet boundaries inside 192.168.1.0/24

A /24 contains 256 addresses, from 192.168.1.0 through 192.168.1.255. Valid /26 boundaries occur every 64 addresses; /27 boundaries occur every 32; and /28 boundaries occur every 16. Subnet allocations must start on the correct boundary and must not overlap.

Example subnet	Network	Usable range	Broadcast	Usable hosts
192.168.1.0/26	192.168.1.0	192.168.1.1–192.168.1.62	192.168.1.63	62
192.168.1.64/27	192.168.1.64	192.168.1.65–192.168.1.94	192.168.1.95	30
192.168.1.96/28	192.168.1.96	192.168.1.97–192.168.1.110	192.168.1.111	14

These three example allocations consume 64 + 32 + 16 = 112 addresses. The remainder is not one continuous block of arbitrary size; it must be allocated using valid network boundaries. The example is only a demonstration of boundary calculation, not a claim about the organisation's current deployment.

Question 3 – Worked VLSM Plan and Efficiency

3.3 Example departmental requirements

For a worked design, assume the departments require 100, 50, and 20 host addresses respectively. These are illustrative figures because the assessment does not give actual requirements. VLSM allocates the largest requirement first so that larger aligned blocks are reserved before smaller subnets are placed.

Department (example)	Required hosts	Smallest suitable subnet	Usable capacity
Department A	100	/25 (128 total)	126
Department B	50	/26 (64 total)	62
Department C	20	/27 (32 total)	30

The three blocks consume 128 + 64 + 32 = 224 addresses, which fits within the 256-address /24. The total usable capacity is 126 + 62 + 30 = 218 host addresses, against 170 assumed required hosts. This leaves 48 usable host slots across the allocated subnets, while 32 addresses remain outside these three allocations for future use or another subnet.

3.4 Recommended allocation

Department	Allocated subnet	Network address	Usable host range	Broadcast
A – 100 hosts	192.168.1.0/25	192.168.1.0	192.168.1.1–192.168.1.126	192.168.1.127
B – 50 hosts	192.168.1.128/26	192.168.1.128	192.168.1.129–192.168.1.190	192.168.1.191
C – 20 hosts	192.168.1.192/27	192.168.1.192	192.168.1.193–192.168.1.222	192.168.1.223
Reserved	192.168.1.224/27	192.168.1.224	192.168.1.225–192.168.1.254	192.168.1.255

3.5 Calculate unused capacity

Department A has 126 usable addresses and 100 are required, leaving 26 unused host slots. Department B has 62 usable addresses and 50 are required, leaving 12 unused host slots. Department C has 30 usable addresses and 20 are required, leaving 10 unused host slots. In total, the allocated department subnets have 218 usable host slots; 170 are required in this example, leaving 48 unused slots within those subnets. The reserved /27 block has 30 usable host slots available for a future department or expansion.

Department	Usable capacity	Assumed requirement	Unused host slots
A	126	100	26
B	62	50	12
C	30	20	10
Total allocated to A–C	218	170	48

Question 3 – Debugging Errors and Optimisation

3.6 Common allocation mistakes to locate

- Overlapping subnets: two departments are assigned ranges that include some of the same addresses. Overlap can cause routing ambiguity and unpredictable delivery.
- Incorrect boundary: a /26 subnet must start on a multiple of 64 in the final octet, a /27 on a multiple of 32, and a /28 on a multiple of 16.
- Wrong mask on a host: a correct IP address with an incorrect mask can make a host treat remote addresses as local or local addresses as remote.
- Insufficient capacity: the subnet has fewer usable host addresses than the current or forecast number of devices.
- Excessive allocation: a department receives a larger block than it needs, reducing the address space available to other departments.
- Network or broadcast address assigned to a host: for ordinary IPv4 subnets, the first address identifies the network and the last is the broadcast address.
- Missing gateway or incorrect DHCP scope: hosts may obtain an address but be unable to reach other subnets because the gateway, pool, or options are wrong.

3.7 VLSM allocation procedure

- Collect the required number of IP interfaces for every department, including printers, access points, servers, and expected growth.
- Sort the departments from largest to smallest requirement.
- For each department, choose the smallest subnet whose usable capacity meets the requirement. In ordinary IPv4, calculate the needed host bits h so that 2^h – 2 is at least the required host count.
- Allocate each subnet on a correctly aligned boundary, beginning with the largest block. Record the network, mask, first host, last host, and broadcast address.
- Check every pair of subnets for overlap and verify that all allocations remain inside 192.168.1.0/24.
- Reserve space for network infrastructure, management, growth, and special-purpose segments where appropriate.
- Document the address plan and validate it in Packet Tracer with routing and host connectivity tests.

3.8 Why VLSM improves utilisation

Fixed-length subnetting gives every department the same block size even when their requirements differ. VLSM allows a large department to receive a /25, a medium department a /26, and a smaller department a /27, for example. The address space is used in proportion to demand, while still leaving a contiguous reserve where possible. This reduces waste and makes future changes easier to plan.

Conclusion: the exact unused-address count for the real organisation cannot be calculated without its actual department requirements and existing allocations. The example demonstrates a valid, non-overlapping VLSM design and the calculations needed to evaluate a real plan.

Question 4 – IPv6 Address Debugging (Repeated Assessment Task)

Question 4 repeats the IPv6 scenario from Question 2. The same technical analysis applies because the addresses and /64 prefix lengths are identical. This section restates the result in a direct question-and-answer format for completeness.

4.1 Complete notation

Device	Expanded IPv6 address
A	2001:0db8:0000:0000:0000:ff00:0042:8329/64
B	2001:0db8:0000:0000:0001:0000:0000:0001/64

4.2 Network prefix and subnet comparison

For a /64 address, the first four hextets form the network prefix. Both devices have the first four hextets 2001:0db8:0000:0000. Therefore, both belong to 2001:db8:0:0::/64. The 1 that appears in Device B's fifth hextet belongs to the interface-identifier portion under /64, not the network prefix.

Check	Result
Device A prefix	2001:db8:0:0::/64
Device B prefix	2001:db8:0:0::/64
Same subnet?	Yes, according to the configured /64 prefixes
Proven prefix mismatch?	No

4.3 Debugging and optimised configuration

No address change is required solely to correct a prefix mismatch because the supplied prefixes already match. Keep both devices on /64 and confirm that their addresses are unique, their interfaces are enabled, and they are connected to the intended link. If the assignment expects an example of a simplified address pair, 2001:db8:0:0::10/64 and 2001:db8:0:0::20/64 are both within the same /64, assuming these values are available and the subnet is assigned to the lab.

The first test should be local IPv6 reachability. If it fails, inspect ICMPv6 Neighbor Discovery messages. A sender may repeatedly issue Neighbor Solicitations without receiving an Advertisement if the peer is disconnected, on a different VLAN, using an incorrect address, or filtering required ICMPv6 traffic. If neighbor resolution succeeds but ping fails, inspect firewall rules and whether ICMPv6 Echo Request/Reply is allowed.

4.4 Available address count

There are 64 bits after the /64 network prefix, so the mathematical interface-identifier space contains 2■■ = 18,446,744,073,709,551,616 bit patterns. IPv6 has no broadcast address, and its address-use rules differ from conventional IPv4, so it is not correct to subtract two as if this were an ordinary IPv4 subnet. Practical allocation must still observe reserved values, duplicate-address detection, and the network's address-management policy.

Question 4 – Verification Checklist and Conclusion

4.5 Practical verification checklist

Test	What to check	Interpretation
Address configuration	Both addresses and /64 prefix are installed correctly	A typo or wrong prefix can cause reachability problems
Interface state	Interface is enabled and link is up	A down interface prevents communication
Neighbor Discovery	Solicitation and Advertisement are exchanged	Failure may indicate a link, VLAN, address, or ICMPv6 issue
ICMPv6 test	Ping peer's full IPv6 address where permitted	A reply confirms basic bidirectional reachability
Firewall policy	Required ICMPv6 traffic is allowed	Overly broad blocking can break IPv6 functions
Topology/VLAN	Both devices are on the intended segment	Same prefix text does not guarantee same physical link
Duplicate address	No other device is using the same address	Duplicate addressing can produce intermittent failures

4.6 Packet Tracer / Wireshark method

In Packet Tracer, assign the two addresses to end devices connected to the same switch and configure /64 on both interfaces. Check the interface status and use the ping command with the peer's full IPv6 address. Enter Simulation mode and filter for ICMPv6 to follow Neighbor Solicitation, Neighbor Advertisement, Echo Request, and Echo Reply messages. In Wireshark, select the correct interface and use `icmpv6` as a display filter. Observe whether the request leaves the source, whether neighbor resolution succeeds, and whether the response returns.

4.7 Conclusion

The supplied values do not contain a /64 prefix mismatch. Both devices belong to 2001:db8:0:0::/64. The assignment's stated communication failure should therefore be treated as a symptom requiring further evidence, not as proof that the addresses are on different subnets. The correct optimisation is to maintain consistent /64 addressing and fix the underlying link, neighbor discovery, filtering, or topology issue identified during testing.

Key IPv6 terminology

- **Prefix:** the leading bits that identify the IPv6 network.
- **Interface identifier:** the remaining bits in an address, often 64 bits for a /64 network.
- **Neighbor Discovery:** ICMPv6 functions used to discover neighboring nodes and link-layer addresses.
- **Duplicate Address Detection:** a process that helps determine whether an IPv6 address is already in use.

Question 5 – Debugging the Routing Table

The router has the following entries. Users report that packets for 192.168.2.100 are not reaching the intended network.

Destination network	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route (0.0.0.0/0)	10.0.0.1

5.1 Apply longest-prefix match

The destination 192.168.2.100 belongs to the network 192.168.2.0/24 because the first 24 bits match the route's network prefix. The router therefore selects the specific 192.168.2.0/24 entry and forwards the packet toward next hop 10.0.0.3, assuming the route is installed and the next hop is reachable.

Candidate route	Does it match 192.168.2.100?	Decision
192.168.1.0/24 via 10.0.0.2	No	Not selected
192.168.2.0/24 via 10.0.0.3	Yes	Selected: longest matching prefix
0.0.0.0/0 via 10.0.0.1	Yes, as a general fallback	Not selected because a more specific match exists

5.2 Verify destination subnet

The valid host range for 192.168.2.0/24 is 192.168.2.1 through 192.168.2.254. Its network address is 192.168.2.0 and its broadcast address is 192.168.2.255. The destination 192.168.2.100 is a valid host address in that subnet.

Property	Value
Network	192.168.2.0/24
Mask	255.255.255.0
First usable host	192.168.2.1
Last usable host	192.168.2.254
Broadcast	192.168.2.255
Destination	192.168.2.100 – valid host

5.3 What can be concluded?

The supplied table contains a specific route that matches the destination. On the information provided, longest-prefix match is not the error: the expected next hop is 10.0.0.3. However, the table alone does not establish that 10.0.0.3 is alive, correctly connected, or able to reach 192.168.2.0/24. If packets are failing, the next hop, downstream routes, interface state, access-control lists, VLANs, and return path should be checked.

Question 5 – Route Optimisation and Final Analysis

5.4 Debugging procedure

- Inspect the router's active routing table and confirm that 192.168.2.0/24 via 10.0.0.3 is installed, not merely present in a configuration file.
- Verify that 10.0.0.3 is reachable on the directly connected transit network and that the local router has a valid route to the next hop.
- Check the router at 10.0.0.3. It must have a route to 192.168.2.0/24, either as a connected network or through another valid route.
- Check the destination host's IP address and subnet mask and ensure that its default gateway is correct.
- Check the return path. The destination network or host must know how to send replies back to the source network.
- Inspect ACLs, firewall rules, NAT policies if used, interface errors, and link state. These can prevent delivery even when the route selection is correct.
- Use traceroute/tracert and packet capture to identify the hop at which packets stop progressing.

5.5 Route optimisation

A routing table should contain accurate, non-conflicting routes with reachable next hops. The existing entries are already logically ordered by specificity in concept, but routers do not rely on row order; they choose the matching route with the longest prefix. The default route 0.0.0.0/0 is used only when no more specific route matches the destination. There is no evidence in the supplied table that an additional route is needed for 192.168.2.100.

If 10.0.0.3 is incorrect or unavailable, correct the next-hop address to the router that actually leads to 192.168.2.0/24, or repair the transit connectivity. Do not replace the specific /24 route with a default route merely to make the table shorter, because that would send the traffic toward 10.0.0.1 instead of the known specific next hop.

5.6 Direct answers

Question	Answer
Matching route for 192.168.2.100	192.168.2.0/24
Selected next hop	10.0.0.3
Is the destination in the intended subnet?	Yes, it is a valid host in 192.168.2.0/24
What should be checked?	Reachability and correctness of 10.0.0.3, downstream/return routes, interfaces, VLANs, and filters
When is the default route used?	When no more specific matching route exists

Overall Conclusion

This assignment demonstrates a systematic method for identifying addressing and routing faults. In Question 1, the IPv4 workstation address belongs to 192.168.10.64/26 while its configured gateway belongs to 192.168.10.0/26; moving the workstation to a valid unused address in the intended /26 resolves the clear subnet inconsistency. In Questions 2 and 4, both IPv6 addresses share the same /64 prefix, so the stated failure cannot be attributed to a prefix mismatch from the provided data alone. In Question 3, VLSM is the appropriate method for fitting different departmental needs into 192.168.1.0/24, but a real unused-address audit requires actual host counts and current allocations. In Question 5, longest-prefix match selects 192.168.2.0/24 via 10.0.0.3; the next hop and downstream path must be verified if delivery still fails.

References / Lab resources

1. CSA07 Computer Networks, Assessment Tool 3, Annexure A: Debugging & Optimisation Task (assessment brief supplied with this assignment).

2. Cisco Packet Tracer – use the simulation environment to configure hosts and routers and inspect packet forwarding.

3. Wireshark – use packet capture and display filters such as `icmp`, `arp`, `ipv6`, and `icmpv6` to verify protocol behaviour.

4. Addressing calculations in this report are derived from the IPv4 subnet masks and IPv6 prefix lengths specified in the assessment brief.

— End of Assignment —